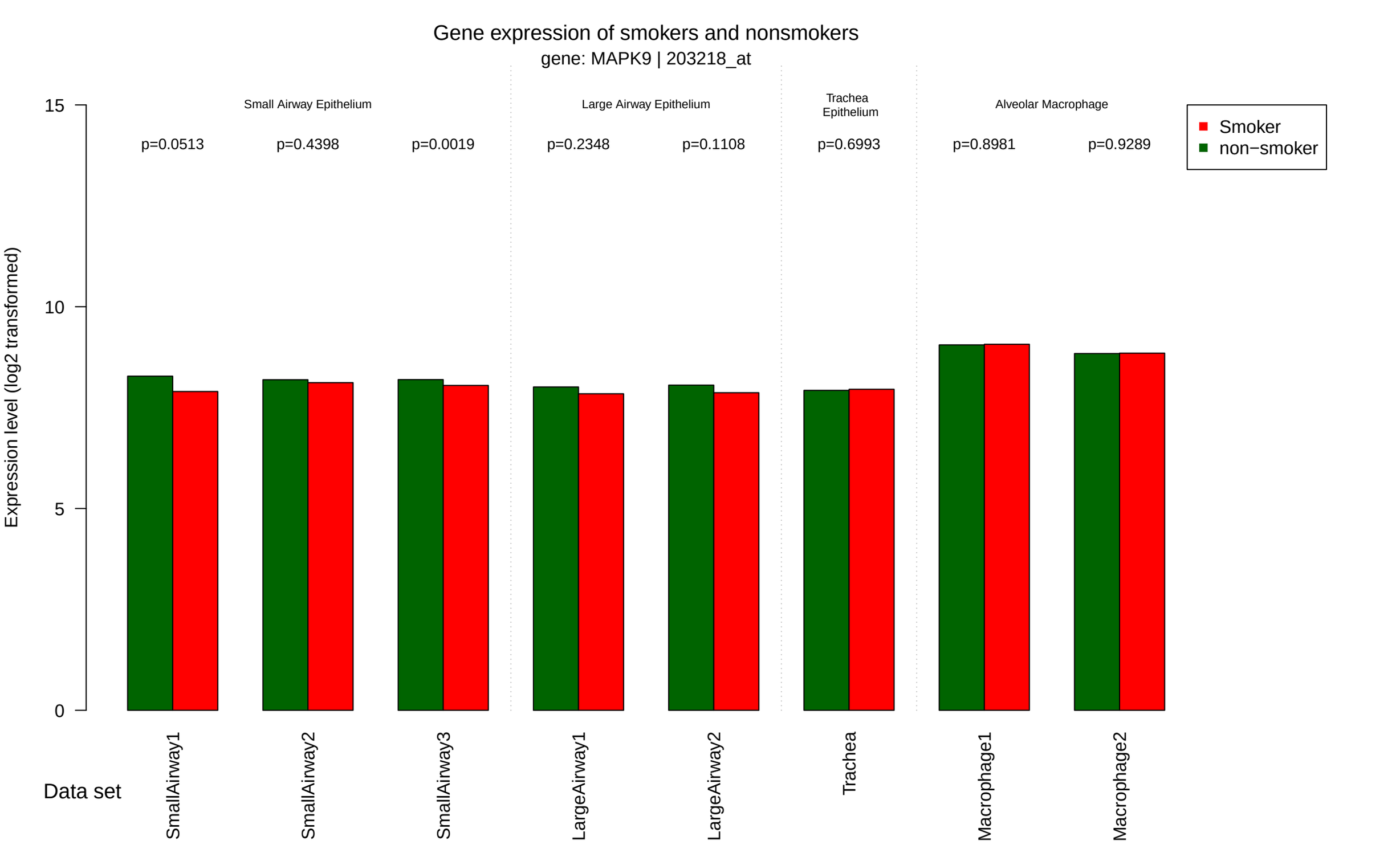
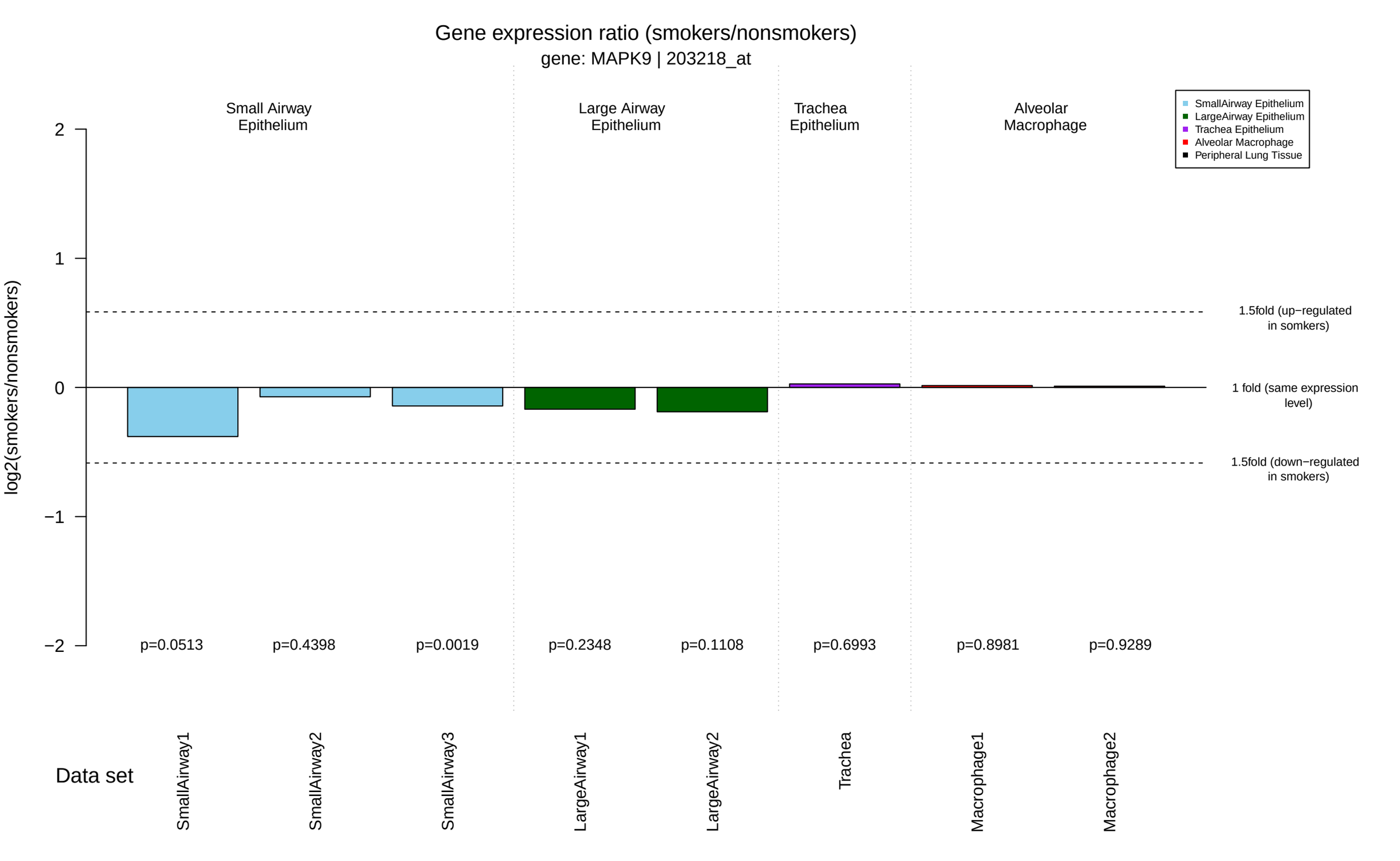
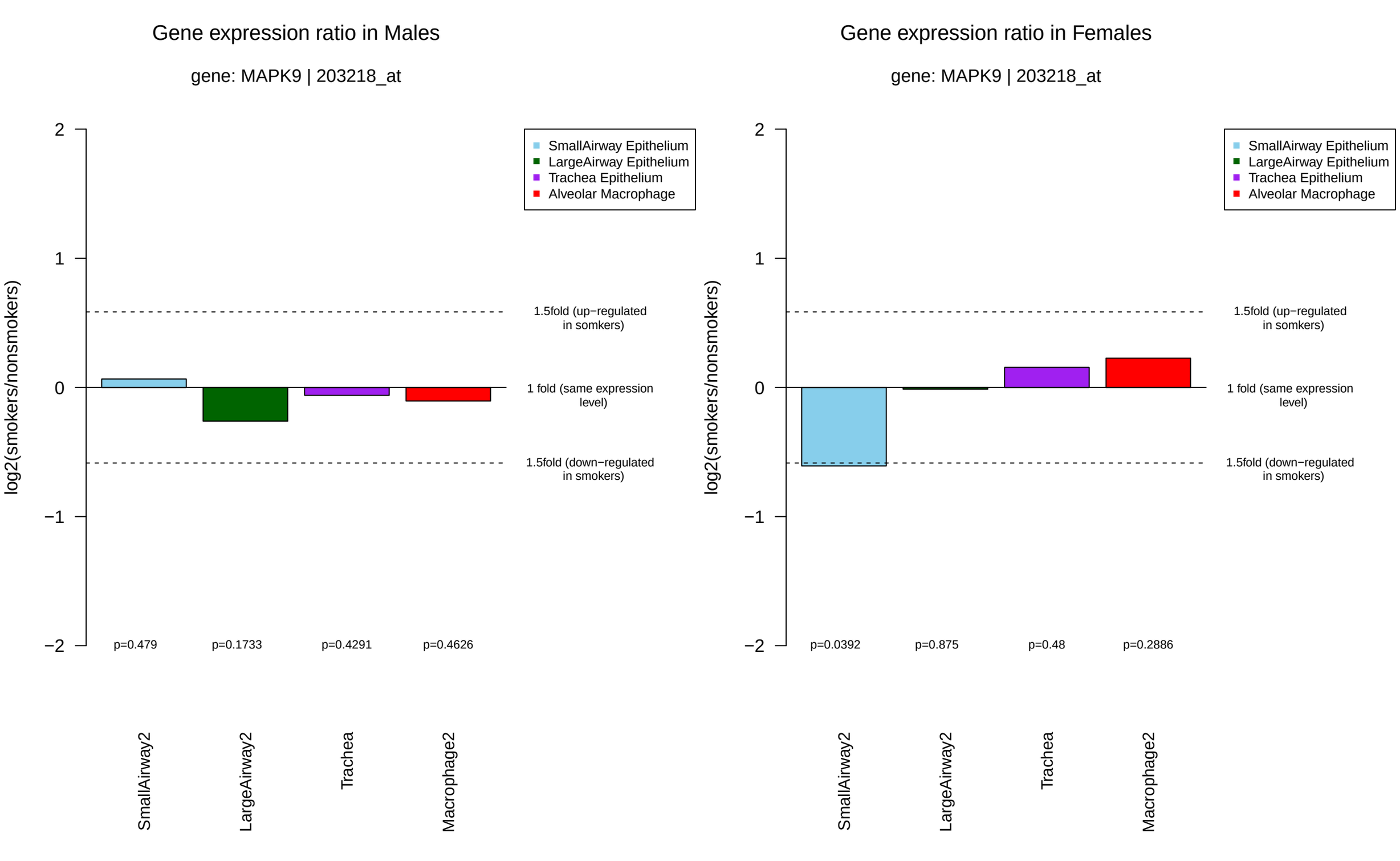
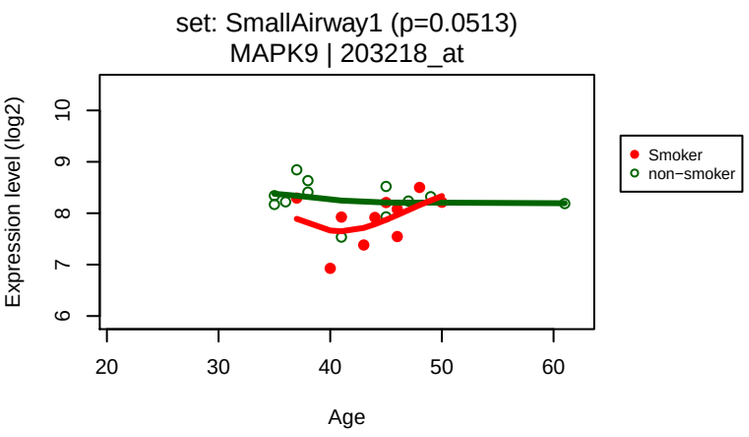
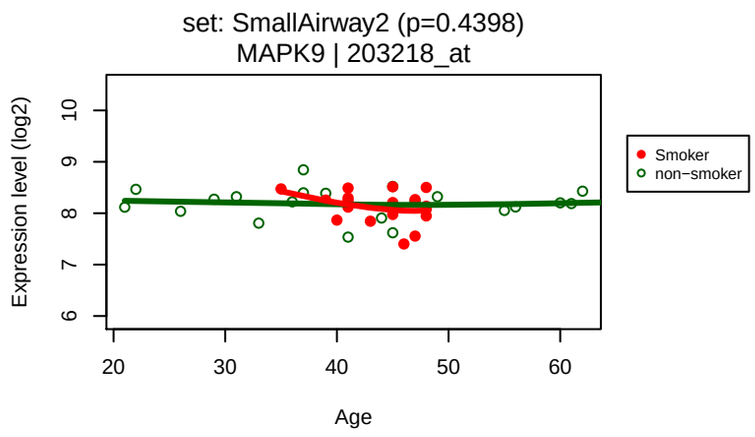
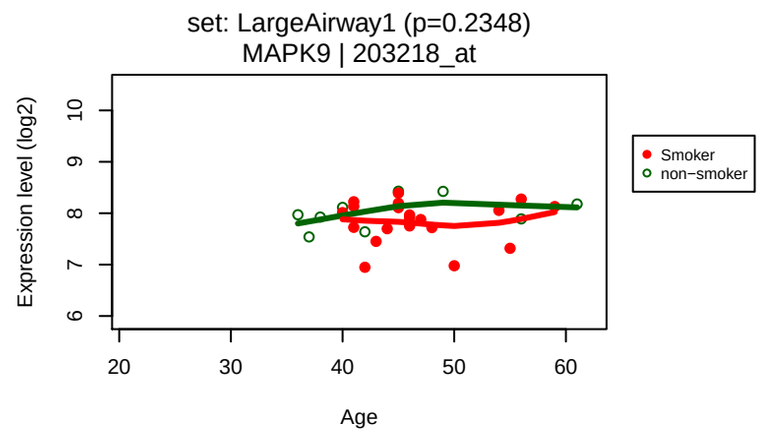

Gene expression of smokers and nonsmokers
gene: MAPK9 | 203218_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0513
p=0.4398
p=0.0019
p=0.2348
p=0.1108
p=0.6993
p=0.8981
p=0.9289
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

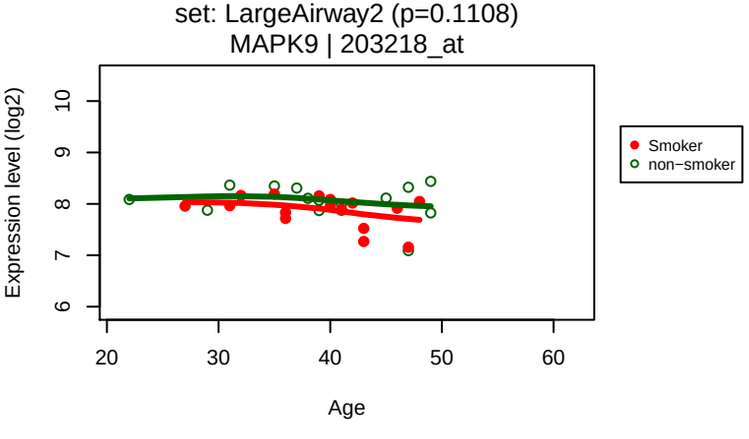

Gene expression ratio (smokers/nonsmokers)
gene: MAPK9 | 203218_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0513
p=0.4398
p=0.0019
p=0.2348
p=0.1108
p=0.6993
p=0.8981
p=0.9289
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

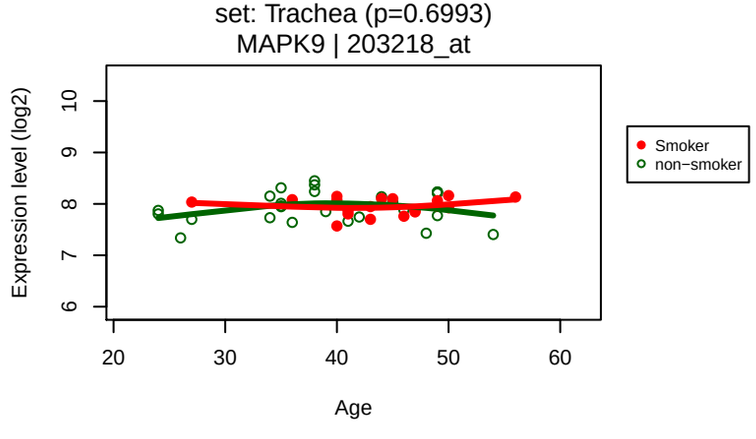

Gene expression ratio in Males
gene: MAPK9 | 203218_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.479
p=0.1733
p=0.4291
p=0.4626
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: MAPK9 | 203218_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0392
p=0.875
p=0.48
p=0.2886
SmallAirway2
LargeAirway2
Trachea
Macrophage2

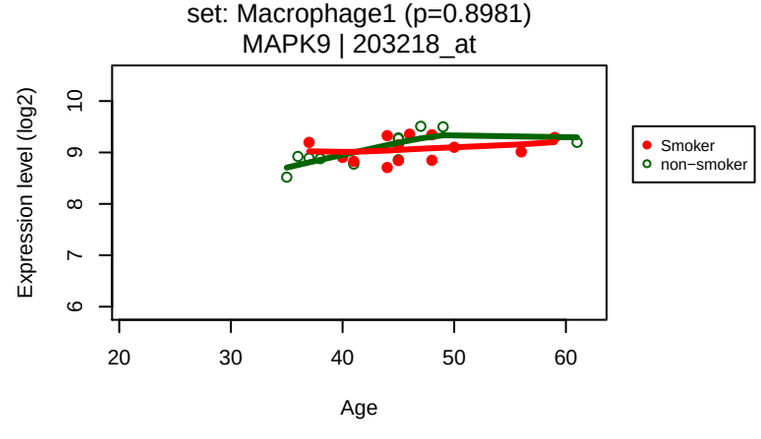

set: SmallAirway1 (p=0.0513)
MAPK9 | 203218_at
Expression level (log2)
Age
Smoker
non−smoker

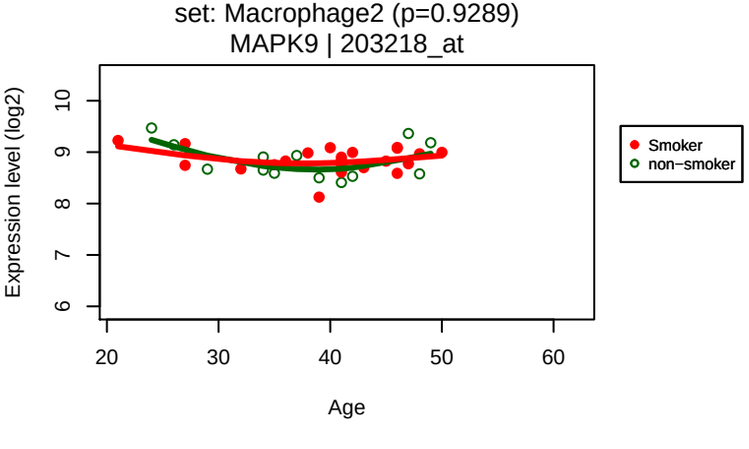

set: SmallAirway2 (p=0.4398)
MAPK9 | 203218_at
Expression level (log2)
Age
Smoker
non−smoker

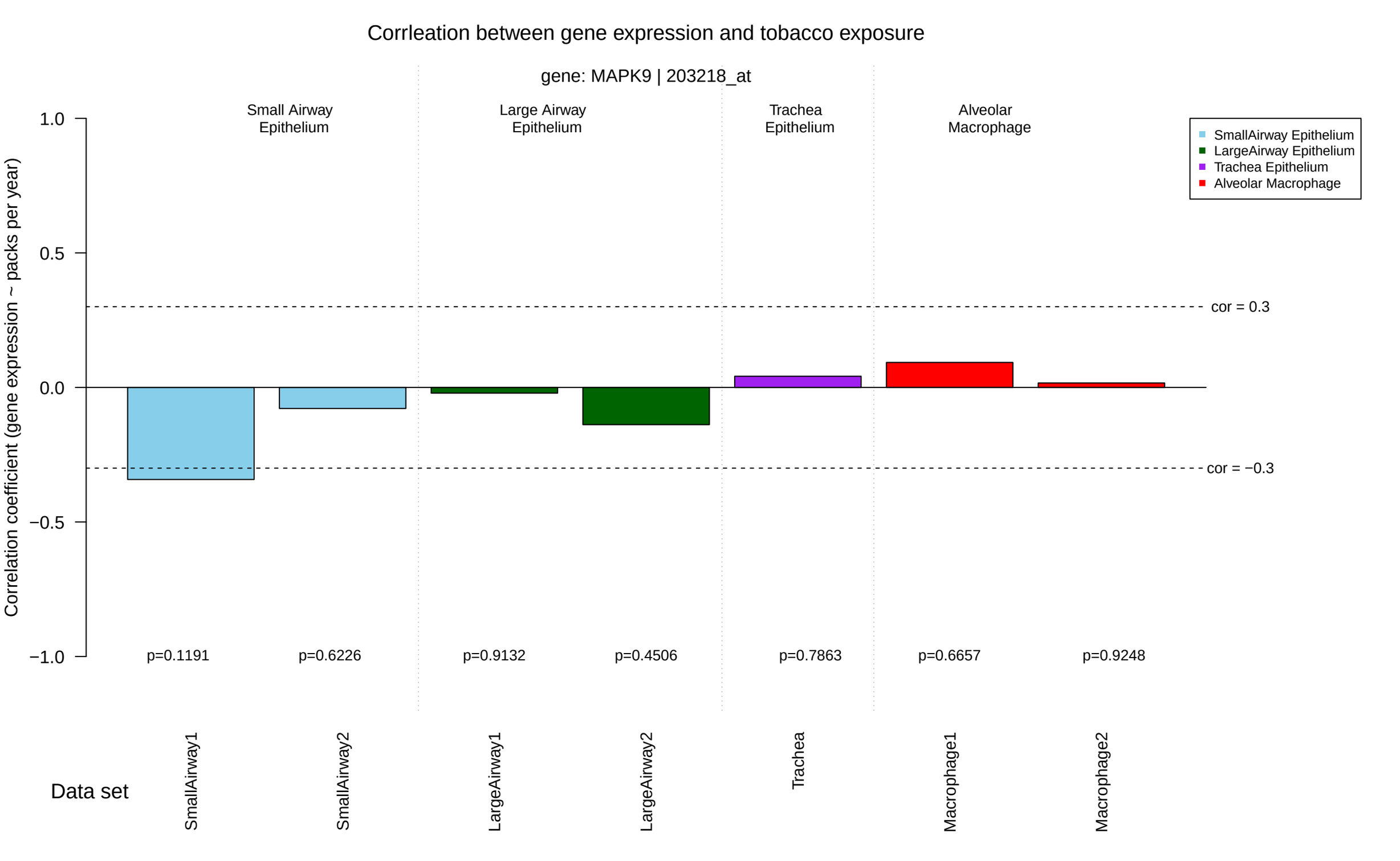

set: LargeAirway1 (p=0.2348)
MAPK9 | 203218_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.1108)
MAPK9 | 203218_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.6993)
MAPK9 | 203218_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.8981)
MAPK9 | 203218_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.9289)
MAPK9 | 203218_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: MAPK9 | 203218_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.1191
p=0.6226
p=0.9132
p=0.4506
p=0.7863
p=0.6657
p=0.9248
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2